

Learning decksh



A scripting language for presentations

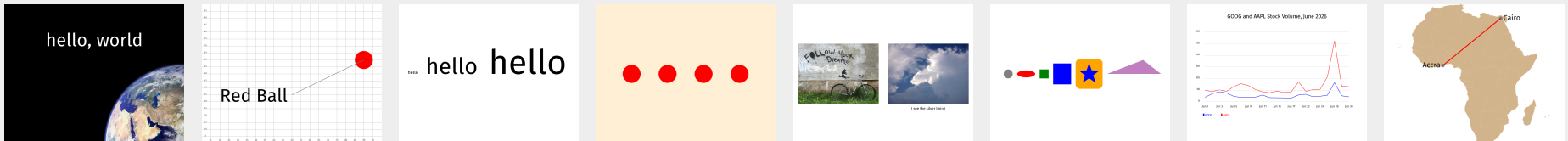
Introduction

decksh is a Domain-Specific Language (DSL) for making presentations, visualizations, and information displays.

decksh works differently from other presentation programs like PowerPoint or Keynote, in that you “program” your deck; that is you build up your slides by specifying high-level objects that represent the elements and attributes of your presentation (text, graphics, lists, charts, maps, etc). The elements are placed and sized using a scalable, percent-based coordinate system, and you can also use variables to describe positions and relationships between the elements.

This tutorial describes installation and basic workflows, and then walks you through various aspects of the language including the coordinate system, the use of variables, colors, text, graphics, images, loops and lists. The tutorial concludes with more advanced topics like charts and maps.

Each concept includes explanatory text, with copy-paste-able examples, and the resulting deck output.



Sections



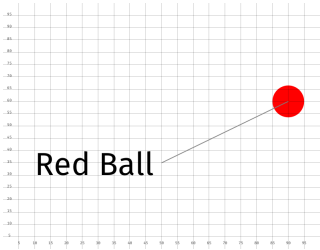
Tools



The Canvas



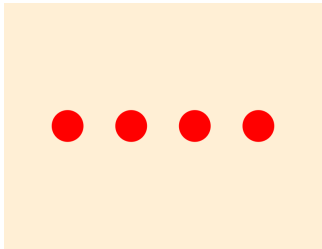
Your First Deck



Coordinates

hello hello

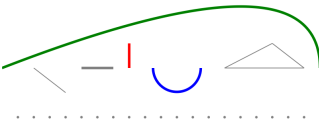
Variables



Colors



Text



Stroked Graphics



Filled Shapes



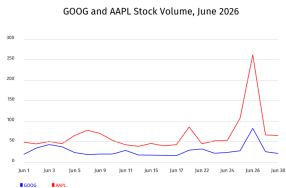
Images



Loops

- First
Second
The last item
- First
 - Second
 - The last item
1. First
2. Second
3. The last item
- First
Second
The last item

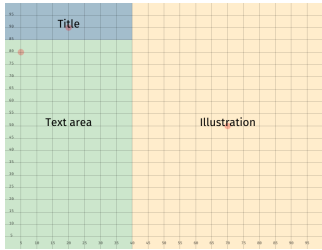
Lists



Charts



Maps



Regions

You have to learn to look at the whole of something, not just the parts.

Gather your tools

In order to make presentations with decksh you will need:

(1) the decksh command-line tool, (2) a text editor to compose and edit the decksh code, (3) a place to execute decksh commands, (4) a PDF viewer to see the results.

See “Installing and Running decksh/pdfdeck” (<https://speakerdeck.com/ajstarks/pdfdeck>) for installation instructions.

Any editor that works with plain text files can be used to compose decksh code. Examples include vim, VS Code, Sublime Text, zed, BBEdit, and micro. All of these editors have syntax highlighting support for decksh, and some editors can automate deck creation.

The general workflow is to compose your decksh code in the text editor, invoke the command to make the result, and then show the results in your PDF viewer. (lines beginning with '\$' are the command line prompt)

```
$ decksh first.dsh > first.xml
```

```
$ pdfdeck first.xml
```

```
$ open first.pdf
```

The example on the right uses Sublime Text for the editor, ghostty for the terminal and Preview as the PDF viewer.



The Slide Canvas

decksh uses a size independent coordinate system, but when designing your slides you should be aware of the canvas size, especially the aspect ratio.

By default, pdfdeck uses a landscape US Letter size (8½ inches high and 11 inches wide), which translates to 792x612 points at 72 points per inch.

Other presets include A3, A4, A5, US Legal (8½ x 14 inches) and Tabloid (11x17 inches). You can also use an arbitrary size such as 1600x900.

Specify the pagesize using the pdfdeck command:

```
$ pdfdeck -pagesize A4 ...
```

```
$ pdfdeck -pagesize 1600x900 ...
```

The shape of the slide canvas is useful when designing your decks. For example a widescreen format like 16:9 is useful for showing multiple images horizontally. Letter and A4 provide a good balance of horizontal and vertical space.



Your First Deck

This is your first deck. Congratulations.

A deck is usually contained in a single text file, conventionally with the suffix ".dsh". Each line in a deck begins with a keyword: a single word that reflects its function. Comments begin with `//` and may be placed at the beginning, or anywhere on a line of decksh code.

The code for the deck is contained between the `'deck'` and `'edek'` keywords. The `'slide'` and `'eslide'` keywords define the boundaries of a slide. You may include an arbitrary number of slides in a single deck.

Keywords are followed by items, known as arguments, separated by spaces or tabs, that define the keyword's behavior. In the example, the `'slide'` keyword has two arguments: "black" and "white", (the slide background is black and its text is white). If you omit the optional arguments, the default is black text on a white background.

The `'ctext'` keyword means centered text. Its arguments are the text ("hello, world"), followed by the coordinates of the text (50% from the left, 70% up from the bottom), followed by the size of the text (10% of the slide width).

The image named "earth.jpg" is centered at the lower right corner (coordinate (100,0)), scaled to be 100% of the slide's width.



```
deck
  slide "black" "white"           // black background, white text
    ctext "hello, world" 50 70 10 // say it loud
    image "earth.jpg" 100 0 100 0 // image in the corner
  eslide
edek
```

The Coordinate System

decksh uses the traditional Cartesian coordinate system to place elements on the canvas. The x, y coordinates represent percentages ranging from 0-100; the coordinates are passed as arguments as a pair of numbers, x first, then y.

The origin (0,0), is at the lower-left, with the x coordinate increasing from left to right, with the y coordinate increasing bottom to top. The middle of the canvas is (50,50): half-way from the left and half-way up from the bottom.

Other measures like the size of text or other objects are represented by the percent of the side width.

For example, to place the text "Red Ball" left-aligned at 10% from the left, and 30% up (10,30), and the text size at 10% of the canvas width, use:

```
text "Red Ball" 10 30 10
```

Other elements like circles and lines may also be placed. For example:

```
circle 90 60 10 "red"
```

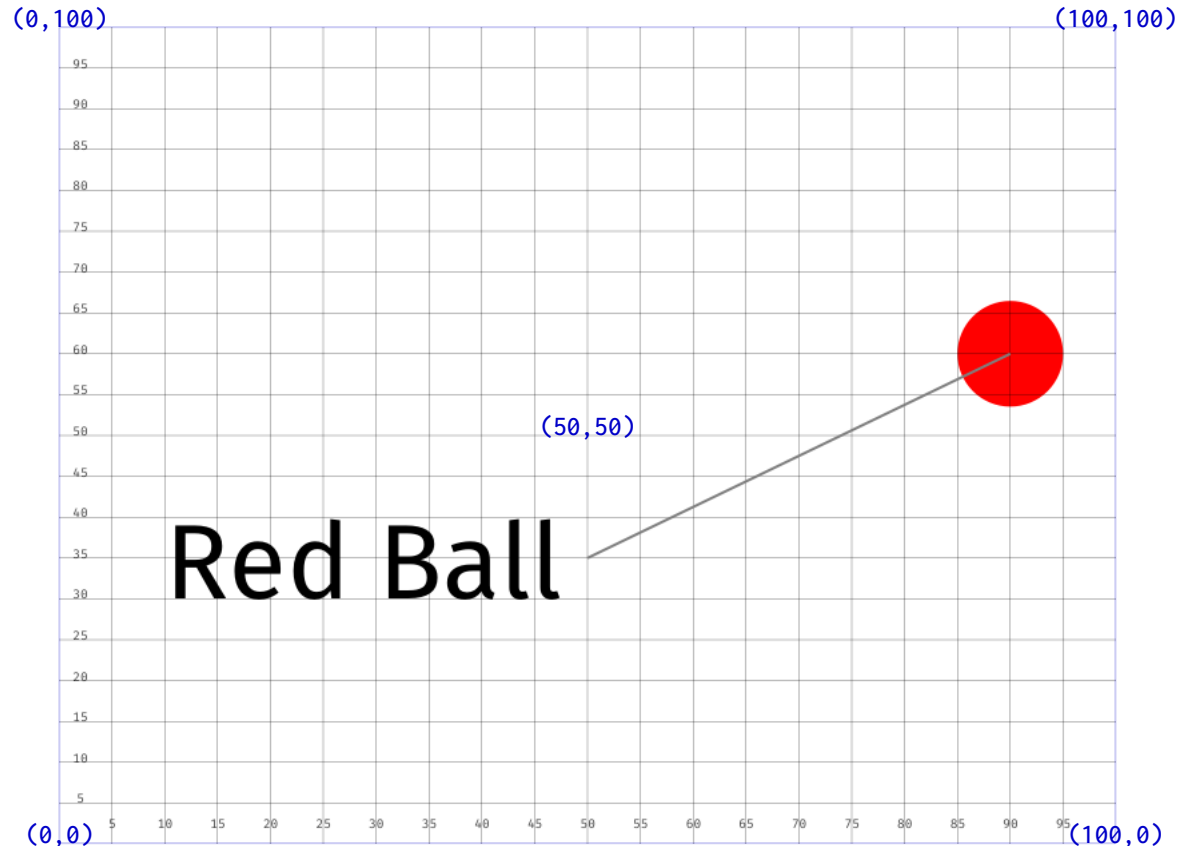
places a red circle at (90,60), the size of the circle is 10% of the canvas width.

To draw a line from (50,35) to (90,60):

```
line 50 35 90 60
```

Finally, note the 'ruler' keyword, which makes a convenient scale.

Place anything, anywhere with precision and consistency.



```
deck
  slide
    ruler                                // show a ruler
    text "Red Ball" 10 30 10             // big text
    circle 90 60 10 "red"                // red circle
    line 50 35 90 60                     // line linking text to circle
  eslide
edeck
```

Variables

Variables provide a method for attaching a name to an object so that the name may be used for other operations.

Variables may refer to either numbers, or strings (text between " characters). Use the '=' after the name to assign the value of variable. The general syntax is:

```
name = value
```

In the example, the value of "hello" is assigned to s, 5 and 50 are assigned to x and y, and ts is assigned the value of 2.5. Next, use the variables:

```
text s x y ts
```

which is equivalent to:

```
text "hello" 5 50 2.5
```

When using variables that contain numbers, you can also perform binary operations: add (+), subtract (-), multiply (*), divide (/), and modulo (%). For example,

```
a = x + y // a gets the sum of x and y
```

You can also use 'assignment operations' on a single variable, for example to add 10 to an existing variable x, use:

```
x += 10
```

Use variables as the levers that control the deck's elements and their relationships.

hello hello

```
deck
  slide
    s="hello"      // s: string to print
    x=5            // x: x coordinate
    y=50           // y: y coordinate
    ts=2.5         // ts: text size
    text s x y ts  // use vars
    x=x+10         // increase x by 10
    ts+=10         // add 10 to text size
    text s x y ts  // use updated vars
    x+=35          // increase x by 35
    ts*=1.5        // text size 1.5
    text s x y ts  // repeat
  eslide
edeck
```


Colors

Many decksh elements use color as an optional argument.

The color may be specified in four ways:

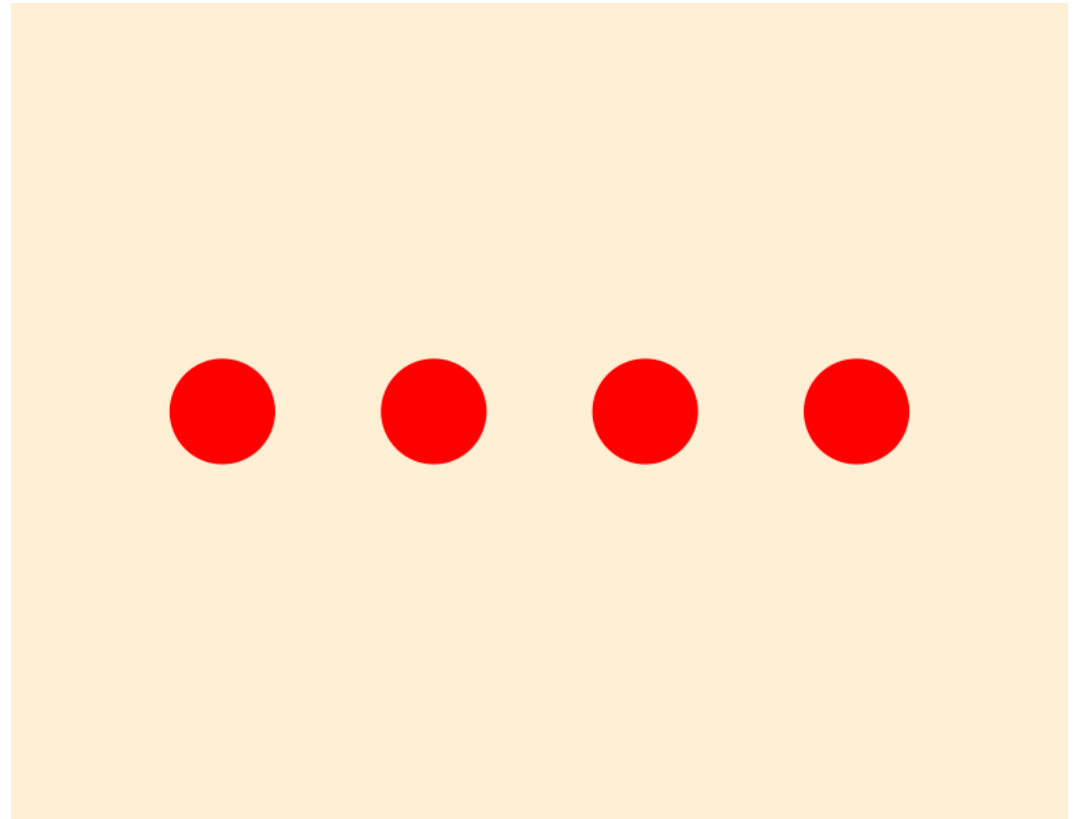
(1) A named color as defined in the web standards; these are colors like "red" and "blue", but also include fairly exotic names like "papayawhip".

(2) A RGB triple, which uses numbers ranging from 0-255 to refer to amount of red, green, and blue in the color. For example, "rgb(0,0,0)" is black, "rgb(255,255,255)" is white, By the way, papayawhip is "rgb(255,239,213)".

(3) The hex RGB format, in the form of "#rrggbb", where rr, gg, and bb are the red, green and blue components of the color as hexadecimal digits ranging from "00" (zero) to "ff" (255). Papayawhip is "#ffefd5" in hex.

(4) The HSV (hue, saturation, value) format uses three numbers: the first, hue, ranges from 0-360; the saturation and value (brightness) range from 0-100.

For example, "hsv(0,100,100)" is a fully saturated red, and "hsv(120,50,50)" is green with diminished saturation and brightness.



```
deck
  slide "papayawhip"           // set slide background color
    c1="red"                   // named color
    c2="rgb(255,0,0)"          // rgb
    c3="#ff0000"               // hex rgb
    c4="hsv(0,100,100)"        // hue, saturation, value
  circle 20 50 10 c1           // red
  circle 40 50 10 c2           // rgb(255,0,0)
  circle 60 50 10 c3           // #ff0000
  circle 80 50 10 c4           // hsv(0,100,100)
eslide
edeck
```

Working with Text

Text is an important decksh element and may be used in various ways. Most text keywords use this pattern:

keyword "text" x y size [font] [color] [op]

Text may be aligned (begin, end, and centered) relative to a specific point. The corresponding keywords are 'text' (and its synonym 'btext') 'etext' and 'ctext'. The point is specified by the x and y coordinate, the size of the text is the third 'size' argument (a percentage relative to the width of the canvas)

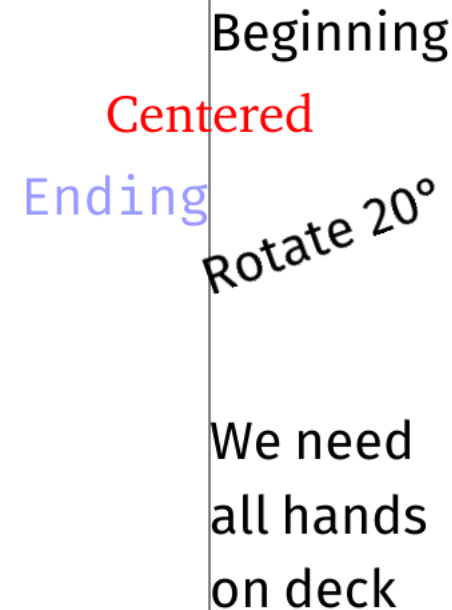
All text keywords have an optional set of arguments that define the font (either "sans", "serif", "mono", or "symbol"), the color of the text, and its opacity (ranging from 0 (invisible) to 100 (solid color)) If no optional arguments are specified the default values are "sans", "black", and 100.

Rotated text ('rtext') is specified like the others, except you must include the angle of rotation (0-360).

rtext "text" x y angle size [font] [color] [op]

A block of text is also similar, but the width of the block must be specified.

textblock "text" x y width size [font] [color] [op]



```
deck
  slide
    tx=50                                     // alignment point
    ts=5                                     // text size
    saying="We need all hands on deck"       // chunk of text
    line          tx 0 tx 100                // midline
    btext "Beginning" tx 90 ts "sans"        // left-aligned
    ctext "Centered" tx 80 ts "serif" "red"  // centered
    etext "Ending"   tx 70 ts "mono" "blue" 40 // end-aligned
    rtext "Rotate 20°" tx 60 20 ts          // rotated
    textblock saying tx 40 20 ts            // block of text
  eslide
edeck
```

Stroked Graphics

decksh supports stroked graphic shapes including lines, polylines, arcs, and curves.

The general syntax includes the arguments for the shape, with optional arguments for line width, color and opacity; (defaulting to 0.2%, "gray" and 100%)

To draw a line between (10,50) and (20,70):

```
line 10 50 20 70
```

Dotted lines are similar, but you may specify the gap between dots after the line width:

```
dline 10 50 90 50 0.2 5
```

Horizontal and vertical lines specify the beginning point and the length:

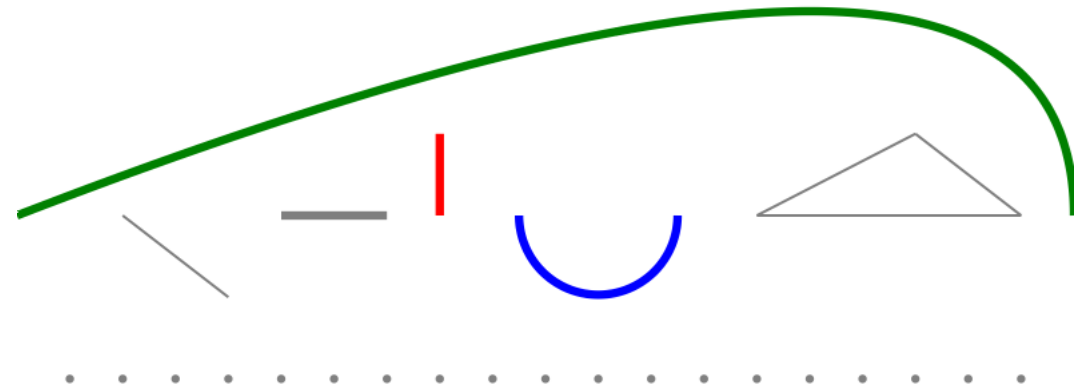
```
hline 10 50 30 and vline 10 50 30
```

For arcs, specify the center of the arc, its width and height, and start and ending angles.

```
arc 50 50 10 10 0 90
```

Polygons use two string arguments: the first is a list of x coordinates, and the second is the y coordinates.

Quadratic Bezier curves take three pairs of coordinates: the beginning point, the control point, and the end point.



```
deck
  slide
    base=50
    lw=0.8
    line    10 base 20 40                // line
    hline   25 base 10 lw                // horizontal line
    vline   40 base 10 lw "red"          // vertical line
    arc     55 base 15 15 180 360 lw "blue" // arc
    polyline "70 85 95" "base 60 base"   // polyline
    curve   0 base 100 100 100 base lw "green" // quadratic bezier
    dline   5 30 95 30 lw 5              // dotted line
  eslide
edeck
```

Filled Shapes

decksh supports filled shapes including circle, ellipse, square, rectangle, rounded rectangle, star, and polygon.

Filled shapes are specified by location arguments (x and y coordinates usually at the center of the shape), followed by dimensions (width and height), with optional arguments for color and opacity; if optional arguments are omitted, the color is "gray" with 100% opacity.

Circles and squares take the center point and a single dimension (width). Ellipses and rectangles also take a center point with a width and height.

Rounded rectangles are specified like regular rectangles, but they require you to specify the radius of the corners.

Filled polygons are specified like polylines: two string arguments that specify the sets of x and y coordinates.

To make stars, specify the center of the star, the number of points, and the inner and outer radii.



```
deck
  slide
    base=50                                     // alignment point
    circle 10 base 5                             // default circle
    ellipse 20 base 10 5 "red"                  // red ellipse
    square 30 base 5 "green"                    // green square
    rect 40 base 10 15 "blue"                   // blue rectangle
    rrect 55 base 10 15 5 "orange"              // orange roundrect
    star 55 base 5 2 6 "blue"                   // blue star
    polygon "65 85 95" "base 60 base" "purple" 50 // purple triangle
  eslide
edeck
```

Images

An image is the welcome companion to text and graphics when making decks.

To place an image, you specify its name (JPEG, PNG, and GIF image formats are supported), location, and dimensions. The general syntax is:

image "name" x y width height

The specified coordinate refers to the middle of the image, and the width and height refer to its dimensions (in pixels).

An alternative (and recommended) syntax scales the image as a percentage of the slide width, preserving the aspect ratio, and you don't have to be concerned with the image dimensions:

image "name" x y canvas-width 0

For example:

image "follow.jpg" 50 50 30 0

Places the JPEG image in the middle of the slide, scaled at 30% of the canvas width.

To add an image caption, use the 'cimage' keyword. The syntax is similar, but an additional caption argument is used:

cimage "cloudy.jpg" "Cloudy Day" 50 50 50 0



I see the silver lining

```
deck
  slide
    image "follow.jpg" 25 50 45 0 // left
    cimage "cloudy.jpg" "I see the silver lining" 75 50 45 0 // right
  eslide
edeck
```

Loops

Loops allow you repeat decksh operations in a controlled way.

A loop is defined as statements bounded by the ‘for’ and ‘efor’ keywords. The ‘for’ arguments define the beginning, ending, and optional increment, which define how many times the statements between ‘for’ and ‘efor’ are executed. The variable specified in the ‘for’ is substituted in the these statements. For example:

```
for v=10 50
```

```
... decksh functions using v
```

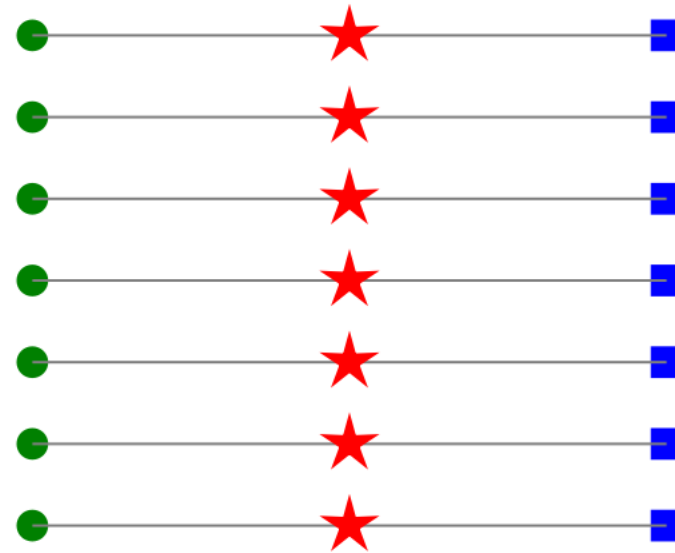
```
efor
```

```
assigns the value ‘v’ from 10 to 50, with an increment of 1.
```

```
for v=10 50 10
```

```
counts from 10 to 50 by 10 (10, 20, 30, 40, 50).
```

In the example on the right, a loop assigns y to go from 10 to 80 by 10, and draws horizontal lines, circles, stars, and squares using the value of y.



```
deck
  slide
    size=3
    for y=20 80 10 // place shapes at the y coordinate 20 to 80 by 10
      hline 20 y 60 // horizontal lines
      circle 20 y size "green" // circles
      star 50 y 5 size 0.8 "red" // stars
      square 80 y size "blue" // squares
    efor
  eslide
edeck
```

Lists

With decksh you can make many kinds of lists: plain, bulleted, numbered, centered.

The type of list is specified by the keyword: 'list' for plain, 'blist' for bulleted, 'nlist' for numbered, 'clist' for centered.

The arguments for lists include the x and y location and the text size of the list.

For example to begin a list at (20, 40) with text size 3 use:

```
list 20 40 3
```

Following the list type is any number of list items (li "item name"), one per line. The list ends with "elist" on a line by itself.

You can construct tables using "parallel lists"; aligning a series of plain lists to a set of columns.

First	● First	1. First	First
Second	● Second	2. Second	Second
The last item	● The last item	3. The last item	The last item

```
deck
  slide
    ls=2.5          // list text size
    ly=95           // top of the list
    colsize=30      // spacing between lists
    x=0             // initial list x coordinate

    list x ly ls     // plain
      li "First"
      li "Second"
      li "The last item"
    elist
    x+=colsize
    blist x ly ls    // bulleted
      li "First"
      li "Second"
      li "The last item"
    elist
    x+=colsize
    nlist x ly ls    // numbered
      li "First"
      li "Second"
      li "The last item"
    elist
    x+=colsize
    clist x ly ls    // centered
      li "First"
      li "Second"
      li "The last item"
    elist
  eslide
edeck
```

Charts

decksh has built-in support for a variety of chart types with these corresponding keywords: barchart, linechart, areachart, scatterchart and dotchart.

To make a chart, specify the chart type and the input file. Data files contain a series of tab separated name,value pairs. For example, a line of data looks like: January<tab>100

Optional arguments control the chart data, label, and value colors. For example to make a line chart from data in "foo.d":

```
linechart "foo.d" [color] [label-color] [value-color]
```

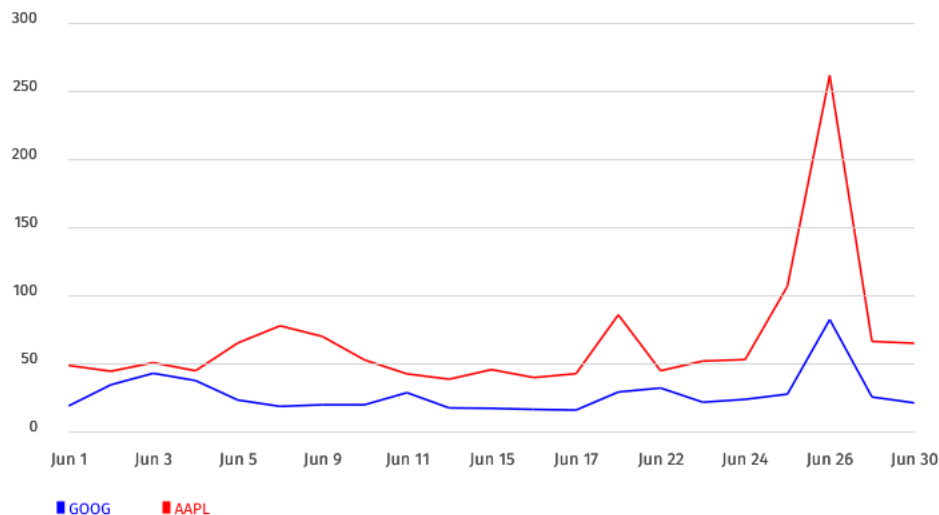
Special variable names control the placement and other aspects of the chart. The variables 'chartTop', 'chartBottom', 'chartLeft', and 'chartRight' define the chart boundaries. If these are not set, the default values are: chartTop=80, chartBottom=30, chartLeft=10, and chartRight=90.

Other variables control items the visibility of values (chartVal), text size (chartTextSize), X-axis label interval (chartXLabel), and Y-axis label's min, max, and interval (chartYRange).

Chart legends are set with the 'legend' keyword using this syntax:

```
legend "text" x y size [font] [color]
```

GOOG and AAPL Stock Volume, June 2026



```
deck
  slide
    ctext "GOOG and AAPL Stock Volume, June 2026" 50 90 3
    chartVal=0 // don't show values
    chartTextSize=2 // set text size to 2%
    chartXLabel=2 // show every other x-axis label
    chartYRange="0,300,50" // set y-axis range and labels

    linechart "aapl-vol.d" "red" // red line chart from AAPL data
    linechart "goog-vol.d" "blue" // blue line chart from GOOG data
    legend "GOOG" 10 20 1.5 "sans" "blue"
    legend "AAPL" 20 20 1.5 "sans" "red"
  eslide
edeck
```


Maps

decksh can read various GIS data files (shapefile, KML, or geoJSON) to produce geographic regions, borders and paths. Plain and labeled locations are also supported.

The keywords for geo functions include 'geoborder' (geographic borders), 'georegion' (regions; useful for countries and continents), 'geopath' (paths between locations), 'geolabel' (a simple label at a location), 'geoloc' (aligned labels ("begin", "center", "end") at a location), and 'geomark' (placing a dot at a location).

To place a map using GIS data in KML format:

```
georegion "africa.kml"
```

Geographic locations may be assigned to variables using geo URL syntax (geo:latitude,longitude), followed by an optional label. For example Los Angeles is defined as:

```
la="geo:34.05,-118.25<tab>Los Angeles"
```

The geographic bounding box of a map is defined by latitude (-90° to 90°) and longitude (-180° to 180°) in decimal degrees.

The 'geobound' keyword maps the specified lat/long boundaries to the slide canvas boundaries that are defined with the 'geocanvas' keyword. The defaults for geobound are -90 90 -180 180, (min,max latitude and min,max longitude) with geocanvas defaulting to 0 100 0 100 (min,max X and min,max Y).



```
deck
  slide
    geobound -34.83 37.34 -17.52 51.266 // boundary of Africa
    geocanvas 15 85 // horiz. scale (15%-85%)
    cairo="geo:30.04440,31.235700 Cairo" // define Cairo, Egypt
    accra="geo:5.614818,-0.205874 Accra" // define Accra, Ghana
    geoloc cairo "b" 4 // label Cairo
    geoloc accra "e" 4 // label Accra
    georegion "africa.kml" "tan" // draw the continent
    geopath accra cairo 0.5 "red" // make path cities
  eslide
edeck
```

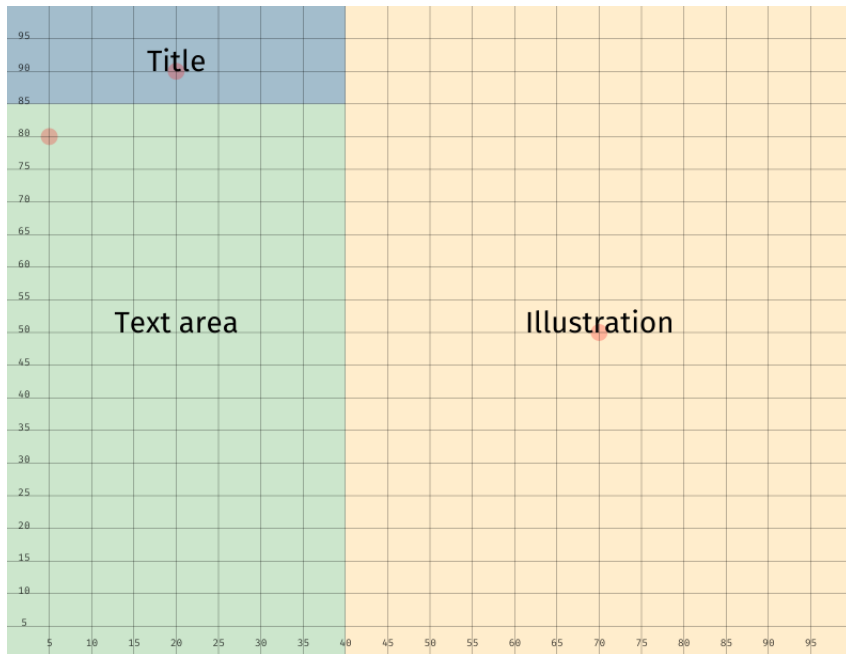
Designing with regions

When designing your slides, it's useful to think in terms of logical regions for your content: A place for titles, text and illustrations for example.

Using decksh variables, you can enforce consistency across slides and a single change will enforce the design throughout the deck.

For example, define the locations of the title (blue), explanatory text (green), and image (orange).

Next use the variables you defined in the initial and subsequent slides.



Using regions ties the deck together.

Why is my cat orange?

New research has shed light on the question of a cat's color ...



```
deck
  tx=20          // title x coordinate
  ty=90          // title y coordinate
  ts=3           // title size
  imx=70         // image x coordinate
  imy=60         // image y coordinate
  ims=50        // image size
  ex=5           // text x coordinate
  ey=80         // text y coordinate
  ew=30         // text width
  es=ts*.75     // text size
slide
  p="New research has shed light on the question of a cat's color ..."
  ctext      "Why is my cat orange?"  tx ty ts
  textblock p                          ex ey ew es "serif"
  image      "cat.png"                imx imy ims 0
eslide
slide
  ctext      "My new cat"              tx ty ts
  textblock "more stuff..."          ex ey ew es "serif"
  image      "newcat.png"             imx imy ims 0
eslide
edeck
```

Building an Example Deck

Now that you understand the fundamentals of making a presentation with decksh, your next step is to exercise your knowledge by making your own deck.

In this example, we will build a presentation on art works in an urban environment.

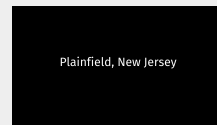
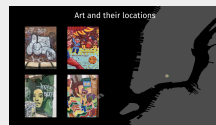
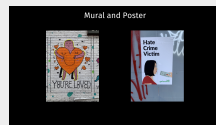
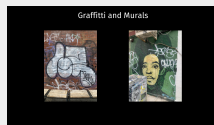
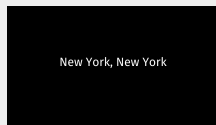
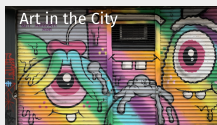
The deck is structured to contain title and section pages, pages that contain two images, pages with an array of images and a map, and image with explanatory text.

Given the design of the deck, the ‘Widescreen’ (16x9in) canvas size is used, along with a “dark mode” color scheme: black background and white text.

The initial code defines a set of variables to be referenced later in the deck.

```
midx=50           // middle x
midy=50           // middle y
hx=30             // hero image x
hy=85             // hero image y
hs=7.5            // hero text size
tx=50             // slide title x
ty=90             // slide title y
ts=hs * 0.5       // slide title size (half hero size)
sx=50             // section x
sy=50             // section y
ss=hs * 0.75      // section text size
limx=30           // left image x
rimx=100-limx     // right image x
imy=50           // image y
ims=25            // image scale
imss=50           // subsection image scale
tbx=60            // textblock x
tby=70            // textblock y
tbw=25            // textblock width
tbs=hs * 0.25     // textblock size (fourth hero size)

bgcolor="black"   // background color
txcolor="white"   // text color
```



Hero and Section

We start the deck with a “hero image” and the title of the deck in white letters. The image is placed in the center, and scaled to use 100% of the canvas width

Next, the slide introducing a section is shown — large white text centered on a black slide.



New York, New York

```
slide // hero image and title
  image "images/chinatown01.jpg" midx midy 100 0
  ctext "Art in the City" hx hy hs "sans" "white"
eslide

slide bgcolor txcolor // section
  ctext "New York, New York" sx sy ss
eslide
```

Two-Up Slides

Graffiti and Murals



Mural and Poster



```
slide bgcolor txcolor // 2-up page
  ctext      "Graffiti and Murals"      tx ty ts
  image      "images/chinatown09r.jpg"   limx imy ims 0
  image      "images/chinatown08r.jpg"   rimx imy ims 0
eslide
slide bgcolor txcolor
  ctext      "Mural and Poster"          tx ty ts
  image      "images/chinatown06r.jpg"   limx imy ims 0
  image      "images/hc.jpg"             rimx imy ims 0
eslide
```

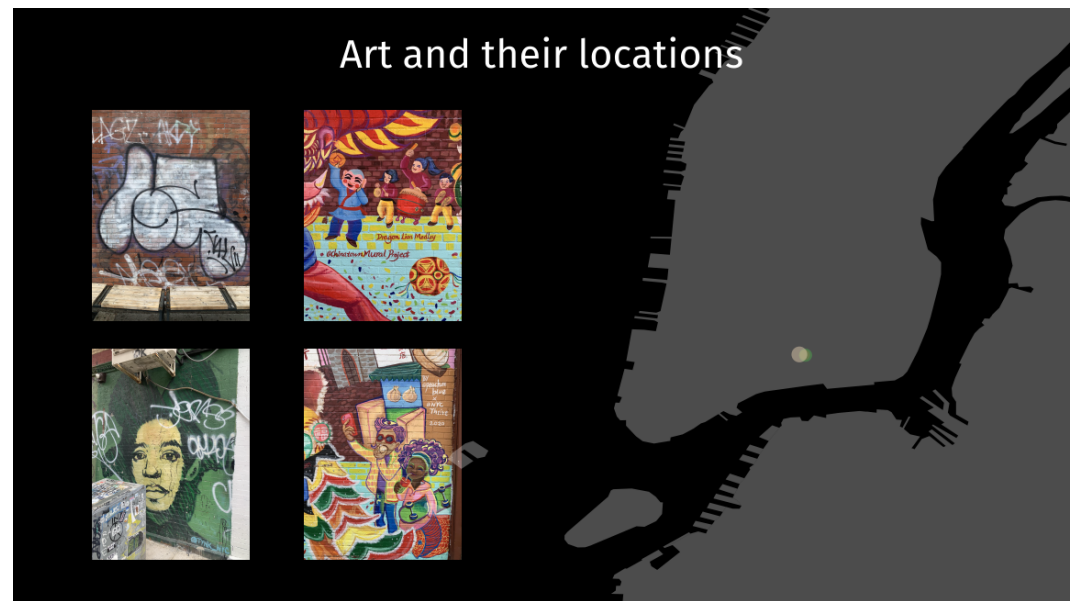

Four Up and Map

This slide is defined by a 2x2 array of images along with their locations on a map.

the bounds of the map (upper left and lower right coordinates) are defined using geo:URL variables, and the image locations (obtained from the image metadata) are also defined.

Next, the slide title is set and the geographic elements are shown.

Finally, the image array is set.



```
slide bgcolor txcolor // 4-up and map
hcolor = "#aa0000"
nw="geo:40.76963076754838,-74.0299466887728" // upper-left of map bound
se="geo:40.69405512120753,-73.96035869194179" // lower-right of map bound

ct01="geo:40.71451100,-73.99131600" // image locations
ct02="geo:40.71462500,-73.99221667"
ct03="geo:40.71454722,-73.99143333"

cctx "Art and their locations" tx ty ts // slide title
geobound se nw // set geographic boundaries
geocanvas 50 95 20 100 // set geo canvas boundaries
georegion "borough.geojson" "white" 30 // show map outline
geomark ct03 1.00 hcolor 50 // mark locations
geomark ct01 1.25 "green" 50
geomark ct02 1.5 "tan" 50

image "images/chinatown04r.jpg" 35 25 15 0 // lower left image
image "images/chinatown03r.jpg" 35 65 15 0 // upper left image
image "images/chinatown08r.jpg" 15 25 15 0 // lower right image
image "images/chinatown09r.jpg" 15 65 15 0 // upper right image
eslide
```

Image and Textblock

Chinatown Murals



The Chinatown Mural Project is a collaboration between activist Karlin Chan and artist Peach Tao. Together, they are bringing back large scale murals to the Chinatown area and beyond. The whimsical and fun culturally appropriate murals depict the culture, history and everyday lives of the community that locals and visitors can relate to.

```
slide bgcolor txcolor // image + text slide
  ctp="The Chinatown Mural Project is a collaboration between activist Karlin Chan and ar
  ctext      "Chinatown Murals"          tx ty ts
  image      "images/chinatown05.jpg"    limx imy imss 0
  textblock ctp tbx tby tbw tbs "serif"
eslide
```

References

[Installing and Running decksh/pdfdeck](#)

[decksh Object Reference](#)

[decksh Keywords](#)

[decksh Visualizations Repository](#)

[decksh Code Repository](#)

How to set up your environment

The definitive decksh language reference

All the keywords and their usage

Repo of decksh examples to learn from

The code for decksh and associated tools and documents